

MI-04: Resolution

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Repository status: Lean verified (original necessity implication). The complete argument received an independent Codex-agent PASS review against the exact catalog target. The original draft was AI-assisted; the reviewed proof below is unchanged. This is independent agent verification, not external human peer review or formal proof certification. [Detailed review and scope](#).

Lean formalization: 16 September 2026

Formalization and verification submission: George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology. Matthew J. Colbrook retains authorship of the mathematical proof below. Substantial OpenAI Codex assistance is disclosed.

The Lean theorem `universal_positive_block_essentially_hermitian` proves the full original necessity implication for every positive finite complex dimension, including singular completions and repeated spectral values. The converse and four-way equivalence in the retained manuscript are outside this formal claim. The twenty-one independent formal targets use the original complex Euclidean operator norm and actual positive-semidefinite matrices.

The verified proof revision is

63340ef17139606dce03c4d9000288129b773157.

The coordinator’s source-bound audit accepted [Linux run 35150473054](#). It authenticates all 233 project inputs, default-kernel replay, all twenty-one Comparator matches, permitted transitive axioms and rejection/isolation controls. An independent nonauthor AI agent also audited the complete execution evidence. The full original necessity implication is Lean verified. Later publication-commit and upstream checks are recorded separately; no future execution is claimed here.

Lean 4.33.1, Mathlib and LeanCert are pinned by the project. The exact certificate $0 < 1/4$ is proved in LeanCert kernel mode and consumed in both required denominator-positivity steps; no interval subdivision is used. The only permitted axioms are `propext`, `Classical.choice` and `Quot.sound`. The source correspondence, reproduction commands and precise verification scope accompany the [immutable Lean source](#). The original mathematical proof is retained verbatim below.

1 MI-04: a characterization of essentially Hermitian off-diagonal blocks

A matrix $X \in \mathbb{M}_n(\mathbb{C})$ is *essentially Hermitian* if $X = \alpha H + \beta I$ for some Hermitian H and complex scalars α, β . The target is the necessity assertion in the following theorem, corresponding to Bourin and Lee’s Conjecture 3.3 [1].

Theorem 1.1. *For $X \in \mathbb{M}_n(\mathbb{C})$, the following conditions are equivalent.*

1. *For every positive semidefinite completion $P = \begin{pmatrix} A & X \\ X^* & B \end{pmatrix}$,*
$$\|P\|_\infty \leq \|A + B\|_\infty.$$

2. The same inequality holds for every such completion whose diagonal-block sum $A + B$ is a scalar multiple of I .
3. For every Hermitian T , the largest and smallest eigenvalues of

$$K_X(T) = \begin{pmatrix} T & X \\ X^* & -T \end{pmatrix}$$

are negatives of one another.

4. X is essentially Hermitian.

In particular, no invertibility or distinct-singular-value hypothesis on X is required.

1.1 Scalar-sum completions force extreme spectral symmetry

The implication (1) $\Rightarrow$ (2) is immediate. Assume (2). Fix a Hermitian T and write

$$a = \lambda_{\max}(K_X(T)), \quad b = -\lambda_{\min}(K_X(T)).$$

Since $\text{tr } K_X(T) = 0$, both a and b are nonnegative. The matrix

$$bI_{2n} + K_X(T) = \begin{pmatrix} bI + T & X \\ X^* & bI - T \end{pmatrix}$$

is positive semidefinite, has diagonal-block sum $2bI$, and has norm $a + b$. Thus (2) gives $a \leq b$.

Let $J = \text{diag}(I, -I)$. Direct multiplication shows

$$K_X(-T) = J[-K_X(T)]J^*.$$

Applying the same argument to $-T$ gives $b \leq a$. This proves (3). If “positive” is interpreted as positive definite, use $b + \eta$ in place of b and let $\eta \downarrow 0$; the same conclusion follows.

1.2 A second-order test forces symmetry of off-diagonal magnitudes

Assume (3). It continues to hold with X replaced by εX , for every $\varepsilon > 0$, because

$$K_{\varepsilon X}(T) = \varepsilon K_X(T/\varepsilon).$$

It is also invariant under unitary similarity of X .

Choose any orthonormal basis and write x_{ij} for the entries of X in that basis. Let

$$D = \text{diag}(1, d_2, \dots, d_n), \quad -1 < d_j < 1,$$

and consider

$$K(\varepsilon) = \begin{pmatrix} D & \varepsilon X \\ \varepsilon X^* & -D \end{pmatrix}.$$

At $\varepsilon = 0$, its largest and smallest eigenvalues are the simple eigenvalues 1 and -1 . For small ε , denote their continuations by $\lambda_+(\varepsilon)$ and $\lambda_-(\varepsilon)$. The elementary second-order perturbation formula gives, with $d_1 = 1$,

$$\lambda_+(\varepsilon) = 1 + \varepsilon^2 \sum_{j=1}^n \frac{|x_{1j}|^2}{1 + d_j} + O(\varepsilon^3), \tag{1}$$

$$\lambda_-(\varepsilon) = -1 - \varepsilon^2 \sum_{j=1}^n \frac{|x_{j1}|^2}{1 + d_j} + O(\varepsilon^3). \tag{2}$$

For completeness, if $H_0 u = \lambda_0 u$ is a simple eigenpair of a Hermitian matrix, $\|u\| = 1$, and V is Hermitian, expand an eigenvector of $H_0 + \varepsilon V$ as $u + \varepsilon w + O(\varepsilon^2)$ with $w \perp u$. In an orthonormal eigenbasis (u_j) of H_0 ,

$$w = \sum_{u_j \perp u} \frac{u_j^* V u}{\lambda_0 - \lambda_j} u_j,$$

and the coefficient of ε^2 in the eigenvalue is $u^*Vw = \sum_{u_j \perp u} |u_j^*Vu|^2/(\lambda_0 - \lambda_j)$. Applying this to the off-diagonal perturbation gives (1)–(2).

Condition (3) implies $\lambda_+(\varepsilon) + \lambda_-(\varepsilon) = 0$. Comparing the quadratic coefficients and canceling the $j = 1$ terms gives

$$\sum_{j=2}^n \frac{|x_{1j}|^2 - |x_{j1}|^2}{1 + d_j} = 0 \quad \text{for every } (d_2, \dots, d_n) \in (-1, 1)^{n-1}. \quad (3)$$

First take all $d_j = 0$. Then change only d_ℓ to any nonzero number in $(-1, 1)$. Subtracting the two identities proves

$$|x_{1\ell}| = |x_{\ell 1}| \quad (\ell \geq 2).$$

The basis was arbitrary, so

$$|u^*Xv| = |v^*Xu| \quad \text{for all orthonormal pairs } u, v. \quad (4)$$

1.3 The pair condition implies normality and a collinear spectrum

Complete a unit vector u to an orthonormal basis and sum the squares in (4). Including the equal diagonal terms yields

$$\|X^*u\|^2 = \|Xu\|^2.$$

Thus $u^*(XX^* - X^*X)u = 0$ for every u , and X is normal.

Use an orthonormal eigenbasis of X , with eigenvalues $z_1, \dots, z_n$. If $n \leq 2$, these eigenvalues lie on a line. For any triple of indices i, j, k when $n \geq 3$, let $\omega = e^{2\pi i/3}$ and set

$$u = \frac{e_i + e_j + e_k}{\sqrt{3}}, \quad v = \frac{e_i + \omega e_j + \omega^2 e_k}{\sqrt{3}}.$$

These vectors are orthonormal. Equation (4) gives

$$|z_i + \omega z_j + \omega^2 z_k| = |z_i + \omega^2 z_j + \omega z_k|.$$

Writing $a = z_i - z_k$ and $b = z_j - z_k$, the difference of the squared sides is

$$|a + \omega b|^2 - |a + \omega^2 b|^2 = 2\sqrt{3} \operatorname{Im}(a\bar{b}).$$

It follows that every eigenvalue triple is collinear. If two eigenvalues are distinct, all the others lie on the line through those two; if none are distinct, X is scalar. Since X is normal, in either case $X = \alpha H + \beta I$ with H Hermitian. This proves (3) $\Rightarrow$ (4).

1.4 Converse and completion of the equivalence

Assume (4). Multiplying X by a scalar of modulus one changes $K_X(T)$ only by unitary similarity. We may therefore write

$$X = H + ibI, \quad H = H^*, \quad b \in \mathbb{R}.$$

With the unitary matrix

$$Q = \frac{1}{\sqrt{2}} \begin{pmatrix} I & I \\ iI & -iI \end{pmatrix},$$

direct multiplication gives

$$Q^* K_X(T) Q = \begin{pmatrix} -bI & T - iH \\ T + iH & bI \end{pmatrix}.$$

A singular value decomposition of $T - iH$ reduces this matrix to n blocks of the form $\begin{pmatrix} -b & \sigma_j \\ \sigma_j & b \end{pmatrix}$. Its eigenvalues are therefore the pairs $\pm \sqrt{\sigma_j^2 + b^2}$. In particular, (3) holds.

Finally assume (3) and let $P = \begin{pmatrix} A & X \\ X^* & B \end{pmatrix} \geq 0$. Set $c = \|A + B\|_\infty$. Because $A + B \leq cI$, the matrix

$$P' = \begin{pmatrix} A & X \\ X^* & cI - A \end{pmatrix} = P + \operatorname{diag}(0, cI - A - B)$$

satisfies $P' \geq P \geq 0$. Also

$$P' = \frac{c}{2}I_{2n} + K_X \left(A - \frac{c}{2}I \right).$$

Positivity gives $\lambda_{\min}(K_X(A - cI/2)) \geq -c/2$. By (3), its largest eigenvalue is at most $c/2$. Thus $P' \leq cI_{2n}$ and consequently $P \leq cI_{2n}$. This proves (1) and finishes the proof of Theorem 1.1. $\square$

References

- [1] Jean-Christophe Bourin and Eun-Young Lee. Eigenvalue inequalities for positive block matrices with the inradius of the numerical range. arXiv:2111.15180, 2021. Conjecture 3.3, printed page 6. URL: <https://arxiv.org/abs/2111.15180>.